Secure By Design

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COURSE 1.5 / COMPANION WORKBOOK

Connect and segment the network

Core routing, default-deny ACLs & internal time

Bring the eight VLANs online through a router the lab controls. Enforce default deny with a documented exception register, stand up the internal time service, and prove every permitted path and denial from the Windows side.

MATERIALS EDITION 1.2 / COURSE 1.5

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NAME _____ / DATE _____

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START HERE / YOUR LEARNING PATH

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Read each lesson alongside the interactive guide and the video. Use the record tables to write down what you actually observe — the workbook is your evidence, not the guide.

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Course review and sign-off

> **HOW TO USE IT** Each lesson page has the same shape: what you will learn, the build steps, the commands that verify them, and a record table. Fill the table from your own environment — never from the sample values.

Before you start

This course connects the eight VLANs through a router the lab controls, enforces default-deny between them with a documented exception register, and stands up the internal time service. The router platform is your choice — the lessons describe what to configure and verify every result from the Windows side, so the checks work whatever appliance you pick. The network still has no path across the air gap.

Prerequisites

- ☐ Complete courses 1.1–1.4: verified domain controllers, DNS, the hardened baseline with its scan cadence, and the two-tier PKI.
- ☐ Provision a new virtual machine for the layer-3 router (SBD-CORE01 in this sample) with one interface per VLAN, and a Windows Server 2025 VM for SBD-TIME01 at 10.0.20.40.
- ☐ Have the Inside the Network VLAN plan at hand: VLANs 10–80 on 10.0.X.0/24 subnets plus management VLAN 100 for network devices, with .1 reserved as each gateway in this sample.

What this course builds

- Design and enforce default-deny segmentation with a justified exception register.
- Build the core router and prove permitted paths and logged denials per VLAN.
- Stand up SBD-TIME01 and verify the domain time hierarchy end to end.

Systems in scope

RECORD	YOUR OBSERVATION
SBD-CORE01 — name, address, OS build	
SBD-TIME01 — name, address, OS build	

LESSON 01 / ABOUT 25 MIN

Design the segmentation and service paths

WHY THIS MATTERS A VLAN is a boundary only if something enforces it. Before touching the router, write the policy: default deny between VLANs, and an exception register where every permitted path names its source, destination, port, and reason.

In this lesson you will

- Translate the Inside the Network VLAN plan into a default-deny segmentation policy.
- Draft the access control list (ACL) exception register from the documented service paths.
- Record the gateway addressing and routing decisions before any device is built.

Follow the build

1. List the eight VLANs with their subnets and purposes from the network plan. Reserve .1 in each subnet as the gateway for this sample, and record that no route crosses the air-gapped boundary.
2. Write the policy sentence first: traffic between VLANs is denied unless a register entry permits it. Segmentation alone does not establish enforcement — the access control lists do.
3. Draft the register's first entries from paths the built systems already need: every VLAN to the domain controllers for DNS, Kerberos, LDAP, SMB, RPC, and time (UDP 123), because domain members take their time from the domain controllers; SBD-DC01, the primary domain controller (PDC) emulator, to SBD-TIME01 — the only system that syncs from the reference clock; and the admin VLAN to the tiers it manages. One row per path: source, destination, service and port, justification, owner.
4. Mark the paths that do not exist yet (workstation pools, business services) as planned rows, and decide the change process: who may add a register entry, and what evidence each addition needs.

Check your result

```
# Record the pre-routing state on each DC — today there is no gateway.
Get-NetIPConfiguration
Get-NetRoute -AddressFamily IPv4 | Select-Object DestinationPrefix, NextHop | Sort-Object
DestinationPrefix -Unique
```

WHAT SUCCESS LOOKS LIKE The register exists with every current path justified, the gateway plan is written, and the pre-routing state of both DCs is captured as the before picture.

Your record

Evidence to keep: Keep the segmentation policy statement, the exception register with owners, and the captured pre-routing output.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	

RECORD	YOUR OBSERVATION
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

What makes a VLAN an enforced boundary?

REMEMBER Default deny between VLANs; the register holds the exceptions. Every permitted path has a source, destination, port, reason, and owner. Gateways are planned (.1 per subnet) and the air gap stays unrouted.

Microsoft · Configure a firewall for Active Directory domains

Microsoft · Get-NetRoute

NIST · SP 800-53 Rev. 5 security controls

LESSON 02 / ABOUT 25 MIN

Build the core router

> **WHY THIS MATTERS** The core is a virtual machine the lab controls: one interface per VLAN, the planned gateway address on each, and no default route anywhere — there is nowhere outside this network to go.

In this lesson you will

- Deploy the layer-3 router VM with an interface in every VLAN.
- Assign the planned gateway addresses with routing enabled and no external route.
- Point the built systems at their gateways and prove basic reachability.

Follow the build

1. Create the router VM (SBD-CORE01 in this sample) with one virtual interface attached to each of the eight VLAN networks. Record the platform and version you chose; the register, not the vendor syntax, is the course's deliverable.
2. Assign each interface its subnet's .1 address and enable routing between the interfaces. Do not configure a default route or any path toward the hypervisor's outside networks — the air gap is a routing decision too.
3. Set the gateway on the built systems: 10.0.10.1 on both domain controllers and 10.0.30.1 on the issuing CA. The offline root gets nothing; it stays off the network entirely.
4. From a host in another VLAN, prove the route: reach each gateway, then a DC address. Leave the ACLs permissive for this one lesson so routing and filtering failures cannot be confused — lesson 3 locks it down.

Check your result

```
Get-NetIPConfiguration | Select-Object InterfaceAlias, IPv4Address, IPv4DefaultGateway
Test-NetConnection 10.0.10.1
Test-NetConnection 10.0.10.10 -TraceRoute
```

> **WHAT SUCCESS LOOKS LIKE** Each built system shows its planned gateway, the trace crosses the router in one hop, and the router holds no route toward anything outside the lab.

Your record

Evidence to keep: Record the router platform and version, its interface and address table, the gateway changes on each system, and the trace output.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	

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RECORD	YOUR OBSERVATION
Evidence file names	
Open items or findings	

Think it through

Why does the router get no default route?

> **REMEMBER** One router interface per VLAN, .1 as each gateway, routing on. No default route — the air gap survives in the routing table. Reachability is proven before any filtering begins.

Microsoft · Test-NetConnection

Microsoft · Get-NetRoute

LESSON 03 / ABOUT 30 MIN

Enforce default-deny with the register

WHY THIS MATTERS Now the policy becomes configuration: deny between VLANs by default, permit exactly the register's rows, and log what gets dropped so the denials are visible evidence, not silence.

In this lesson you will

- Implement default-deny filtering between all VLANs.
- Permit the Active Directory, PKI, and time service paths from the register.
- Log denied traffic so enforcement produces evidence.

Follow the build

1. Apply the inter-VLAN default-deny rule on the router, with logging for denied packets. The log of what was blocked is as much evidence as the list of what was allowed.
2. Permit the directory paths from every VLAN to VLAN 10 per Microsoft's Active Directory port documentation: DNS (53), Kerberos (88 and 464), LDAP (389/636), global catalog (3268/3269), SMB (445), and the RPC endpoint mapper with its dynamic range — the register row for RPC should name the range decision explicitly.
3. Permit time the way Windows Time actually flows: UDP 123 from every VLAN to both domain controllers, and UDP 123 from SBD-DC01 alone to SBD-TIME01. Permit HTTP to the pki.sbd.test publication service in VLAN 30 so clients can fetch certificates and revocation lists.
4. Record the VLAN 80 peer denies now — workstation-to-workstation SMB and Remote Desktop — as host-firewall rows: traffic inside one VLAN never reaches the router, so course 1.9 enforces them with each PC's own firewall. Then read back the full rule set against the register line by line — anything configured but not registered gets removed or registered, never ignored.

Check your result

```
# From a test host outside VLAN 10: permitted paths answer, everything else fails.
Test-NetConnection SBD-DC01.sbd.test -Port 53
Test-NetConnection SBD-DC01.sbd.test -Port 88
Test-NetConnection SBD-DC01.sbd.test -Port 389
Test-NetConnection 10.0.10.10 -Port 3389   # expected: TcpTestSucceeded False
```

WHAT SUCCESS LOOKS LIKE Registered paths succeed, an unregistered port fails and appears in the router's deny log, and the rule set matches the register with no unexplained entries in either direction.

Your record

Evidence to keep: Keep the exported rule set, the register cross-check, the permitted-path test output, and a captured deny-log entry for the blocked test.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	

RECORD	YOUR OBSERVATION
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

A rule exists on the router but not in the register. What happens?

> **REMEMBER** Default deny is in force; the register's rows are the only openings. Denied traffic is logged — enforcement you can show, not assume. Rule set and register reconcile exactly, both directions.

Microsoft · Configure a firewall for Active Directory domains

Microsoft · Test-NetConnection

LESSON 04 / ABOUT 35 MIN

Stand up SBD-TIME01 and the time hierarchy

> **WHY THIS MATTERS** Kerberos tolerates minutes of skew, not more, and an air-gapped network has no internet time to lean on. Build the internal reference, point the PDC emulator at it, and let the domain hierarchy carry time to everything else.

In this lesson you will

- Configure SBD-TIME01 as the lab's internal reference clock.
- Point the PDC emulator at it and keep the rest on the domain hierarchy.
- Document the accuracy, fallback, and drift decisions the model calls for.

Follow the build

1. Build SBD-TIME01 at 10.0.20.40 as a workgroup or domain-joined Windows Server 2025 host with the Windows Time service configured as an NTP server. In an air-gapped lab its clock is the reference by declaration — record the source decision (host clock, or an approved hardware source) with its expected accuracy and drift threshold.
2. On SBD-DC01, the domain controller holding the PDC emulator role, configure Windows Time to use SBD-TIME01 as its manual peer and mark it reliable. Only the PDC emulator points at TIME01: SBD-DC02 syncs from the PDC emulator, every other domain member keeps the default domain-hierarchy sync from the domain controllers, and the domain-joined Linux servers in later courses point chrony or ntpd at both domain controllers.
3. Confirm the register's time rows are active — members to both domain controllers on UDP 123, and SBD-DC01 alone to SBD-TIME01 — and that the hardening Group Policy objects (GPOs) from course 1.3 do not conflict with the time settings. If a Security Technical Implementation Guide (STIG) value and this design disagree, the deviation record from that course is where the decision lives.
4. Verify the chain end to end: TIME01 serving, the PDC emulator sourcing from it, SBD-DC02 and a member server sourcing from a domain controller, and offsets within your recorded threshold.

Check your result

```
Get-ADDomain | Select-Object PDCEmulator   # expected: SBD-DC01.sbd.test
# On the PDC emulator, then on SBD-DC02 and a member server:
w32tm /query /source
w32tm /query /status
w32tm /stripchart /computer:10.0.20.40 /samples:3 /dataonly   # from SBD-DC01 only
```

> **WHAT SUCCESS LOOKS LIKE** The PDC emulator reports SBD-TIME01 as its source, SBD-DC02 and the member server report a domain controller, and measured offsets sit inside the documented threshold.

Your record

Evidence to keep: Record the time design decisions, the source and status output from both controllers and the member, the stripchart offsets, and the register rows that permit time: members to the domain controllers, and SBD-DC01 to SBD-TIME01.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

Which system syncs directly with SBD-TIME01?

> **REMEMBER** TIME01 is the declared reference; its accuracy decisions are written down. Only the PDC emulator peers with it; the hierarchy carries the rest. Offsets are measured and inside the threshold, not assumed.

Microsoft · Windows Time service

Microsoft · Windows Time service tools and settings

LESSON 05 / ABOUT 25 MIN

Prove the paths and the denials

WHY THIS MATTERS A segmented network is proven twice: every registered path works from where it is supposed to work, and the paths you never granted fail — visibly, in the log.

In this lesson you will

- Test the permitted-path matrix from each VLAN against the register.
- Prove representative denials and capture their log entries.
- Confirm the directory still behaves correctly across the routed network.

Follow the build

1. From a test host in each populated VLAN, run the register's permitted paths as a scripted matrix: directory ports and time to VLAN 10, and HTTP to pki.sbd.test. Record source VLAN, destination, port, and result for every row.
2. Run the deliberate failures: a port no register row permits, from a VLAN that should not reach it — for example, time straight to SBD-TIME01 from a test host outside VLAN 20, since only SBD-DC01 may reach it. Confirm the failure and find its entry in the router's deny log — the pair is the evidence.
3. Re-run the course 1.2 and 1.4 verification habits across the boundary: resolve the domain SRV records and fetch pki.sbd.test's certificate revocation list from outside VLANs 10 and 30.
4. File the matrix and the deny-log captures with the register. Any mismatch — a registered path failing or an unregistered path passing — is a finding to resolve before the course ends, not a note.

Check your result

```
$targets = @(@('SBD-DC01.sbd.test',53),@('SBD-DC01.sbd.test',88),@('SBD-DC01.sbd.test',389),@('pki.sbd.test',80))
foreach ($t in $targets) { Test-NetConnection $t[0] -Port $t[1] | Select-Object ComputerName, RemotePort, TcpTestSucceeded }
w32tm /stripchart /computer:SBD-DC01.sbd.test /samples:3 /dataonly   # time over UDP 123 answers
w32tm /stripchart /computer:10.0.20.40 /samples:1 /dataonly   # from a test host outside VLAN 20: expected to fail (only SBD-DC01 may reach TIME01)
Resolve-DnsName _ldap._tcp.dc._msdcs.sbd.test -Type SRV
```

WHAT SUCCESS LOOKS LIKE Every register row passes from its source VLAN, every probed unregistered path fails with a logged deny, and directory discovery works across the routed network.

Your record

Evidence to keep: Keep the test matrix with timestamps, the time queries, the deny-log captures, and the cross-VLAN SRV and CRL retrieval output.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	

RECORD	YOUR OBSERVATION
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

What is the complete evidence for one denial?

> **REMEMBER** Permits proven per VLAN; denials proven with their log entries. Directory discovery and CRL retrieval work across the boundary. Register and reality reconcile before the course moves on.

Microsoft · Configure a firewall for Active Directory domains

Microsoft · Test-NetConnection

LESSON 06 / ABOUT 25 MIN

Protect the router's own management

WHY THIS MATTERS The router enforces every boundary, so whoever manages it controls them all. Give its management a separate network, a short list of who may reach it, a log of every change, and a versioned copy of every configuration.

In this lesson you will

- Move the router's management interface onto a dedicated management VLAN.
- Restrict management access to the Tier 0 administration seat and secure protocols.
- Log router changes and keep versioned, hashed configuration backups.

Follow the build

1. Create management VLAN 100 (10.0.100.0/24) for network-device management only, and bind the router's SSH and HTTPS management to its address there, 10.0.100.1. Disable Telnet, plain HTTP, and every management service the platform enables by default that the design does not use.
2. Permit management sessions to 10.0.100.1 only from AdminJumpBox01's planned address (10.0.60.10), which course 1.7 builds — until then, manage from the hypervisor console only. Add the row to the register: the router enforces Tier 0 boundaries, so its management is Tier 0.
3. Give each administrator a named account on the router and keep one sealed break-glass account; shared logins hide who changed what. Course 1.8 sends the router's syslog, including its deny log, to the collector; write that as a dated follow-up in the register now.
4. Export the running configuration after every approved change, hash it, and keep every version. Compare the newest export with the running configuration on a schedule; a difference with no change record is a finding.

Check your result

```
# From a host outside VLAN 100 (expected: TcpTestSucceeded False):
Test-NetConnection 10.0.100.1 -Port 22
# Configuration versions with hashes:
Get-FileHash -Algorithm SHA256 C:\Evidence\connect-segment\router-config-*.txt
```

WHAT SUCCESS LOOKS LIKE Management answers only on VLAN 100 and only from the permitted source, insecure management services are off, and each configuration version has a hash and a matching change record.

Your record

Evidence to keep: Keep the management-access rule export, a refused connection test from a user VLAN, the list of disabled services, and the hashed configuration versions with their change records.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	

RECORD	YOUR OBSERVATION
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

Why does the router's management get its own VLAN?

> **REMEMBER** Router management lives on VLAN 100, reachable from the Tier 0 seat only. Insecure management services are off; logins are named, with one sealed break-glass account. Every configuration version is hashed and tied to a change record.

Microsoft · Test-NetConnection

NIST · SP 800-53 Rev. 5 security controls

LESSON 07 / ABOUT 15 MIN

Delta scan the change and hand off

WHY THIS MATTERS Course 1.3's rule was simple: every role addition gets a scan. Two systems joined the network and every existing one gained a gateway — that is a change worth measuring.

In this lesson you will

- Run the change-triggered scans on the touched Windows systems.
- Capture and hash the router configuration as its own baseline.
- Hand the register and evidence into the standing cadence.

Follow the build

1. Rescan both domain controllers with the same benchmark and tool versions as course 1.3, and take SBD-TIME01's first baseline scan. Attribute any movement to this course's changes — the gateway and time settings are the expected deltas.
2. Export the router's configuration, hash the export, and store it with the evidence. The router has no Security Content Automation Protocol (SCAP) benchmark in this lab, so its config export, register reconciliation, and deny log stand as its assessable record — say so explicitly in the evidence notes.
3. Add the new rows to the hardening course's exception and deviation record where the two overlap (time settings, any firewall interplay), and set the register's named owner and review date alongside the scan cadence.
4. Write the handoff line for the next course: the network now routes, filters, and keeps time — course 1.6's PKI publication host and template administration build on exactly these paths.

Check your result

```
Get-ChildItem C:\Evidence\connect-segment -Recurse | Select-Object Name, LastWriteTime
Get-FileHash -Algorithm SHA256 C:\Evidence\connect-segment\*
```

WHAT SUCCESS LOOKS LIKE The DC rescans show only the expected deltas, TIME01 has a dated baseline, the router export is hashed and reconciled with the register, and owners and review dates are set.

Your record

Evidence to keep: Finish the course checklist: scan results, router export and hash, the reconciled register with owner and review date, and the handoff note.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	

RECORD	YOUR OBSERVATION
Evidence file names	
Open items or findings	

Think it through

Why does this course end with scans at all?

REMEMBER Change-triggered scans ran; deltas are attributed. The router's record is its hashed export plus the register and deny log. Owners, review dates, and the handoff to course 1.6 are written.

NIST · SP 800-53 Rev. 5 security controls

Course review and sign-off

Before moving on, confirm each lesson left evidence behind — and that what remains open is written down, not remembered.

Lesson completion

LESSON	EVIDENCE FILED	DATE
01 Design the segmentation and service paths	☐ yes	
02 Build the core router	☐ yes	
03 Enforce default-deny with the register	☐ yes	
04 Stand up SBD-TIME01 and the time hierarchy	☐ yes	
05 Prove the paths and the denials	☐ yes	
06 Protect the router's own management	☐ yes	
07 Delta scan the change and hand off	☐ yes	

Open items carried forward

Next in the learning path

That completes course 1.5. Course 1.6 finishes the PKI services: the pki.sbd.test publication site on SBD-PKI-MGMT01 and certificate template administration.

> **KEEP IT HONEST** Lesson completion records learning activity; it is not a system authorization decision. Your observations, not the sample values, establish what happened in your environment.

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